Fractions: Prime Factorizations

Factors of Whole Numbers

The **factors of a whole number** are numbers that divide into it evenly (no remainder). Below are the factors of 18.

Factors of 18: 1, 2, 3, 6, 9, 18

The factors of a number can be used to write the number as a product. Below, the number 18 has been written as a product using the factors 6 and 3.

$$18 = 6 \cdot 3$$

Note that we can also write the number 18 as a product using the factors 2 and 9.

Problem 1 *List all factors of the number 20.*

Multiple Choice:

(a) $1, 2, 3, 4, 5, 10, 20$

(b) $1, 2, 4, 5, 10, 20$ ✓

(c) $1, 2, 4, 5, 10, 15, 20$

(d) $1, 2, 4, 5, 10$

Hint: *Write your own list of factors, rather than trying to find the answer by looking at the options. This will help you to understand and remember the process.*

Problem 2 *List all factors of the number 24.*

Multiple Choice:

(a) $1, 2, 3, 4, 6, 8, 24$

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(b) $1, 2, 3, 4, 6, 12, 24$

(c) $1, 2, 3, 4, 6, 8, 12, 24$ ✓

(d) $2, 3, 4, 6, 8, 12$

Problem 3 *List all factors of the number 36.*

Multiple Choice:

(a) $1, 2, 3, 4, 9, 12, 18, 36$

(b) $1, 2, 3, 4, 6, 8, 9, 12, 18, 36$

(c) $1, 2, 3, 4, 6, 9, 12, 18, 36$ ✓

(d) $1, 2, 3, 4, 6, 6, 9, 12, 18, 36$

Feedback(*`attempt`*): ***Note****: While $6 \times 6 = 36$, we only list the number 6 once when writing out the set of factors.*

Prime and Composite Numbers

Definition 1 (**Prime and Composite Numbers**)**.** A **prime number** is a **natural number** greater than 1 that is evenly divisible *only* by itself and 1. Equivalently, a **prime number** is a **natural number** greater than 1 whose only factors are 1 and itself.

The first several prime numbers are 2, 3, 5, 7, 11, 13, 17, 19, 23 and 29.

A natural number greater than 1 that's not prime is called a **composite number**.

Procedure 1 (**How to Tell if a Number Less Than 100 is Prime**)**.** To check if a number less than 100 is prime, you only need to test if it can be evenly divided by **2, 3, 5, or 7**. If none of these four numbers work, the number is prime.

Rather than performing long division, we can use the **divisibility rules** to check if a number is divisible by 2, 3, or 5.

Fact 1. The divisibility rules for 2, 3 and 5 are as follows:

- A number is divisible by 2 if it is even.

- A number is divisible by 3 if the sum of its digits is divisible by 3.
- A number is divisible by 5 if it ends in 0 or 5.

Example 1. Are the numbers 23 and 35 prime or composite?

Explanation. We need to check if 23 can be evenly divided by 2, 3, 5, or 7.

- $23 \div 2 = 11.5$ (not evenly) - $23 \div 3 = 7.67$ (not evenly) - $23 \div 5 = 4.6$ (not evenly) - $23 \div 7 = 3.29$ (not evenly)

Since none of these four numbers divide into 23 without a remainder, the number is prime.

Problem 4 *Which of the following numbers are* ***prime****?*

2, 9, 15, 19, 27, 31

Select All Correct Answers:

(a) *2* ✓

(b) *9*

(c) *15*

(d) *19* ✓

(e) *27*

(f) *31* ✓

Problem 5 ***True or False*** *All prime numbers are odd, except for 2.*

Multiple Choice:

(a) *True* ✓

(b) *False*

Feedback(*`correct`*): *That's right! Since all even numbers are evenly divisible by 2, the only one that can be prime is 2.*

Prime Factorization

The **prime factorization** of a composite number is the number written as the product of prime numbers. For example, the prime factorization of $18 = 2 \cdot 3 \cdot 3$. We will use **factorization trees** to find the prime factorization of a composite number.

Procedure 2 (Finding the Prime Factorization of a Number). To find the prime factorization of any composite number, you can create a **factorization tree** using the steps below.

Step 1 **Find a Pair of Factors**: Identify two numbers that have a product equal to the composite number. Draw two branches extending from the composite number to the factors.

Step 2 **Evaluate Each Factor**: If a factor is prime, circle it. This branch is now finished. If a factor is composite, do not circle it. Continue to the next step.

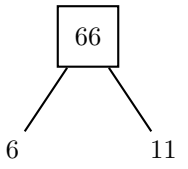
Step 3 **Repeat the Process**: For every number that is not circled, find a pair of factors and draw new branches.

Step 4 **Final Result**: Once every branch ends in a circled prime number, you are finished. Write the prime factorization by multiplying all the circled numbers together.

Example 2. Find the prime factorization of the number 66.

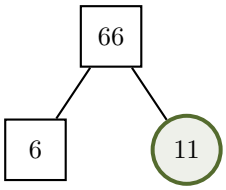
Remark 1. In the figures below, each composite number is written inside a square for clarity. You can omit the squares when finding prime factorizations by hand, but don't forget to circle the prime numbers!

Explanation. Step 1 **Find a Pair of Factors**: The numbers 6 and 11 have a product of 66.



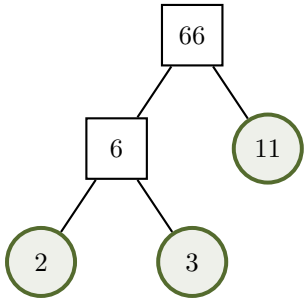
Alt text: A partial factorization tree for the number 66. The top node is 66, which branches into 6 and 11.

Step 2 **Evaluate Each Factor**: Since 11 is prime it's circled. The number 6 is not prime, so the process is repeated in the next step.



Alt text: A partial factorization tree for the number 66. The top node is 66, which branches into 6 and 11. The number 11 is circled since it's prime.

Step 3 **Repeat the Process**: We can write 6 as the product of 2 and 3. Since both 2 and 3 are prime, they are circled and the tree is now finished.

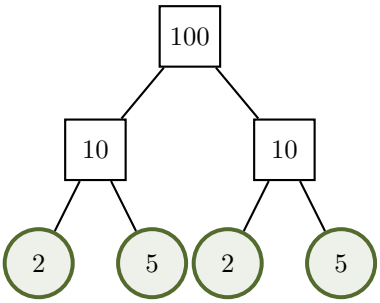


Alt text: A factorization tree for the number 66. The top node is 66, which branches into 6 and 11. The number 11 is circled since it's prime. The number 6 branches further into 2 and 3, both of which are circled since they are prime.

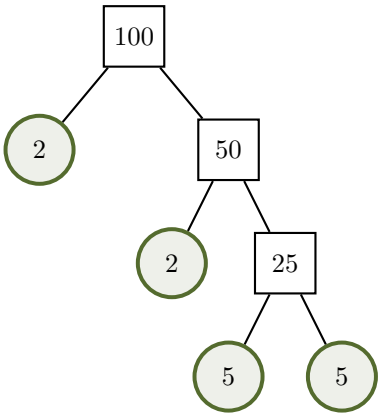
Step 4 **Final Result**: The prime factorization of 66 is the product of all circled numbers in the tree.

$$66 = 2 \cdot 3 \cdot 11$$

Remark 2. No matter which factors you choose to start with, your tree will always end with the same set of prime numbers. In the figure below, there are two factorization trees for 100. One begins with the factors 10 and 10 and the other begins with 2 and 50, but they both result in the same prime factorization.



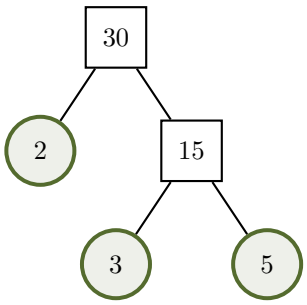
$100 = 2 \cdot 2 \cdot 5 \cdot 5$



$100 = 2 \cdot 2 \cdot 5 \cdot 5$

Alt text: Two factorization trees for the number 100 shown side-by-side. The left tree branches into two 10s, each branching into prime factors 2 and 5. The right tree branches into 2 and 50, then 50 branches into 2 and 25, then 25 branches into 5 and 5. Both trees show the final prime factorization 100 equals 2 times 2 times 5 times 5.

Problem 6 *The factorization tree for the number 30 is given below.*



Alt text: A factorization tree for the number 30. The top node is 30, which branches into 2 and 15. The number 2 is circled. The number 15 branches further into 3 and 5, both of which are circled.

Based on this tree, what is the prime factorization of 30*?*

Multiple Choice:

(a) $2 \cdot 15$

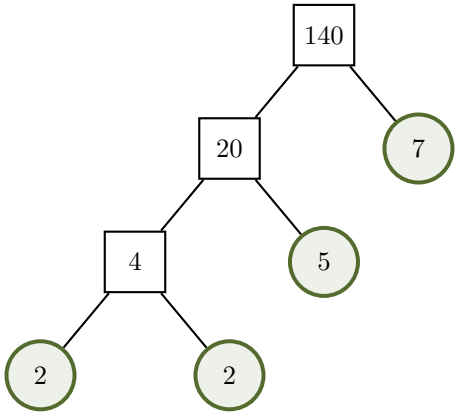
(b) $2 \cdot 3 \cdot 5$ ✓

(c) $2 \cdot 3 \cdot 10$

(d) $2 \cdot 3 \cdot 5 \cdot 15$

Feedback(`attempt`): *To find the prime factorization, only multiply the numbers in the circles at the very ends of the branches. Composite numbers in boxes (like 15) are just stepping stones.*

Problem 7 *The factorization tree for the number 140 is given below.*



Alt text: A factorization tree for the number 140. The top node is 140, branching into 20 and 7. The 20 branches into 4 and 5, and the 4 branches into two circled 2s. The prime numbers 2, 2, 5, and 7 are circled.

Based on this tree, what is the prime factorization of 140*?*

Multiple Choice:

(a) $2 \cdot 2 \cdot 35$

(b) $4 \cdot 5 \cdot 7$

(c) $2 \cdot 2 \cdot 5 \cdot 7$ ✓

(d) $2 \cdot 5 \cdot 7 \cdot 20$

Problem 8 *Find the prime factorization of* 42*.*

Multiple Choice:

(a) $6 \cdot 7$

(b) $2 \cdot 21$

(c) $2 \cdot 3 \cdot 7$ ✓

(d) $2 \cdot 3 \cdot 3 \cdot 7$

Hint: *Think of a pair of factors for* 42. *For example,* $42 = 6 \cdot 7$. *Since* 7 *is prime, you only need to factor* 6.

Problem 9 *Find the prime factorization of* 120.

Multiple Choice:

(a) $10 \cdot 12$

(b) $2 \cdot 2 \cdot 3 \cdot 10$

(c) $2 \cdot 2 \cdot 2 \cdot 3 \cdot 5$ ✓

(d) $2 \cdot 3 \cdot 4 \cdot 5$

Extra Insight: The Power of Prime Numbers

Did you know that **prime numbers** keep your online data safe from hackers? Secure messages and accounts are locked by a giant composite number and the key is its prime factorization. One way this is done is through a process called **RSA Encryption**. To create the lock, a computer picks two very large prime numbers and multiplies them together to get an enormous composite number. A computer can multiply two very large prime numbers in less than a second to create the composite number, but finding its prime factorization is very difficult! It would take the fastest supercomputer billions of years. Since its nearly impossible for hackers to find the key (the prime factorization), your accounts and messages stay locked and your information remains private!

This whole process depends on knowing two very large prime numbers. The largest known prime number was discovered in 2024 and is over 41 million digits long! People use a network of thousands of computers (called GIMPS) to hunt for these massive prime numbers and there is even a cash prize for the person who finds the next one.